A Global Structural Reduction for the Spherical-Shell Proxy

Global analytic structural module

Abstract

This note gives the global structural component of the purely analytic middle-ratio argument for the spherical-shell proxy. The weighted proxy is first written as an exact layer-cake sum. Its deficit from the action integral is then decomposed into a radial shifted-quadrature deficit and an angular rounding defect. A Stieltjes calculation retains two nonnegative remainders that are discarded by the optical estimate, and one of those remainders admits a smooth explicit lower bound. The result is the concrete sufficient inequality (37). All structural identities hold for every $0 < \rho < 1$ and $K > 0$; in particular they apply on the upward-closed middle-ratio region $\rho_c \le \rho \le 39/50,\ K \ge k_{11}(\rho)$.

1 Action and the global target

For $a > 0$ and $z \ge 0$, put

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \le z < a, \\ 0, & z \ge a. \end{cases} \tag{1}$$

Fix $0 < \rho < 1$ and $K > 0$, and define

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

The strict positive-integer count is

$$[x]_< := \#\{n \in \mathbb{N} : n < x\}.$$

The spherical-shell proxy and its continuous payment are

$$P(\rho, K) := \sum_{\substack{\nu = \ell + 1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \left[A(\nu) + \tfrac{1}{4}\right]_<, \tag{3}$$

$$W(\rho, K) := \int_0^K 2z A(z)\, dz = \frac{2(1-\rho^3)K^3}{9\pi}. \tag{4}$$

The upward-closed middle-ratio region used below is

$$\mathcal{R}_c := \left\{(\rho, K) : \rho_c \le \rho \le \frac{39}{50}, \quad K \ge k_{11}(\rho)\right\}, \qquad \rho_c := \frac{1}{1 + 2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}. \tag{5}$$

The middle-ratio target is $P < W$ on $\mathcal{R}_{\mathrm{c}}$.

For $0 < z < K$, direct differentiation gives

$$-A'(z) = \frac{1}{\pi}\begin{cases} \arccos\dfrac{z}{K} - \arccos\dfrac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos\dfrac{z}{K}, & \rho K \le z < K. \end{cases} \tag{6}$$

In particular, A is strictly decreasing from T to 0. Let

$$R : [0,T] \longrightarrow [0,K]$$

be its decreasing inverse, and set

$$F(t) := R(t)^2, \qquad g(t) := -F'(t), \qquad \tau := A(\rho K) = KG_1(\rho). \tag{7}$$

The derivative in (6) increases in magnitude on $(0, \rho K)$ and decreases in magnitude on $(\rho K, K)$. The inverse curvature has the related, but slightly stronger, property

$$g \text{ decreases on } (0,\tau), \qquad g \text{ increases on } (\tau, T). \tag{8}$$

Indeed, on the outer branch, if $z = R(t)$ then

$$g(t) = \frac{2\pi z}{\arccos(z/K)};$$

the quotient increases with z, while z decreases with t. On the inner branch, put

$$\delta(z) := \arccos\frac{z}{K} - \arccos\frac{z}{\rho K}.$$

Differentiation with respect to the intermediate ratio gives

$$\frac{\delta(z)}{z} = \int_\rho^1 \frac{ds}{s\sqrt{s^2K^2 - z^2}}.$$

This integral increases with z. Thus $z/\delta(z)$ decreases with z, and $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint and interface data are

$$F(\tau) = \rho^2K^2, \qquad g(\tau) = \frac{2\pi\rho K}{\arccos\rho}, \qquad g(T) = \frac{2\pi\rho K}{1-\rho}. \tag{9}$$

Here and below $g(T)$ is the one-sided limit. Near $t = 0$, one has $g(t) = O(t^{-1/3})$, so all improper Stieltjes boundary terms used below vanish.

2 Exact layer cake and angular rounding

Let

$$x_n := n - \frac{1}{4}, \qquad M(r) := \#\left\{\ell \in \mathbb{N}_0 : \ell + \frac{1}{2} < r\right\} = \max\left\{0, \left\lceil r - \frac{1}{2}\right\rceil\right\}. \tag{10}$$

Lemma 1 (Exact layer cake)**.** *For every $0 < \rho < 1$ and $K > 0$,*

$$P(\rho, K) = \sum_{\substack{n\ge 1\\ x_n < T}} M(R(x_n))^2, \qquad W(\rho, K) = \int_0^T F(t)\,dt. \tag{11}$$

Both identities retain the strict convention at every radial and angular wall.

Proof. For a fixed half-integer ν,

$$\left[A(\nu) + \frac{1}{4}\right]_< = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since A and R are decreasing, $x_n < A(\nu)$ is equivalent to $\nu < R(x_n)$. Interchanging the two finite sums and using

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2$$

proves the first identity. At $R(x_n) = m + 1/2$, the strict angular count is m, exactly as specified by M.

For the second identity, the area formula for a decreasing inverse gives

$$\int_0^T R(t)^2\, dt = \int_0^K 2zA(z)\, dz.$$

The last integral is (4). □

For every active n, abbreviate

$$r_n := R(x_n), \qquad m_n := M(r_n), \qquad \alpha_n := r_n - m_n + \frac{1}{2}. \tag{12}$$

The strict convention implies

$$m_n - \frac{1}{2} < r_n \leq m_n + \frac{1}{2}, \qquad 0 < \alpha_n \leq 1. \tag{13}$$

This includes $m_n = 0$. Define the angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{x_n < T} (m_n^2 - r_n^2). \tag{14}$$

The defect of a single radial layer is exactly

$$m_n^2 - r_n^2 = (1 - 2\alpha_n) r_n + \left(\alpha_n - \frac{1}{2}\right)^2. \tag{15}$$

Unlike the one-sided estimate $m_n^2 - r_n^2 < r_n + 1/4$, formula (15) retains the cancellation when $\alpha_n > 1/2$.

Combining (11) and (14) gives the first exact two-term decomposition:

$$W - P = \mathcal{D}_{\text{rad}} - \mathcal{E}_{\text{ang}}, \qquad \mathcal{D}_{\text{rad}} := \int_0^T F(t)\, dt - \sum_{x_n < T} F(x_n). \tag{16}$$

3 The radial deficit with retained positive remainders

Define the shifted counting function and its two primitives by

$$C(t) := \#\{n \geq 1 : x_n < t\}, \quad w(t) := t - C(t), \quad \mathfrak{W}(t) := \int_0^t w(s)\, ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \tag{17}$$

On $0 \le t \le 3/4$,

$$\Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \tag{18}$$

For $t = x_n + u$, $0 \le u \le 1$,

$$\Psi(t) = \frac{1}{2}\left(u - \frac{1}{2}\right)^2 - \frac{1}{32}. \tag{19}$$

Consequently

$$\mathfrak{W}(t) \ge 0, \qquad -\frac{1}{32} \le \Psi(t) \le \frac{3}{32} \quad (t \ge 0). \tag{20}$$

Proposition 2 (Exact retained-remainder identity)**.** *For every* $0 < \rho < 1$ *and* $K > 0$*, put*

$$\mathcal{R}_{\mathrm{dec}} := -\int_{(0,\tau)} \mathfrak{W}(t)\, dg(t), \tag{21}$$

$$\mathcal{R}_{\mathrm{osc}} := \left(\Psi(T) + \frac{1}{32}\right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi(t)\right) dg(t). \tag{22}$$

Then both remainders are nonnegative, and

$$\begin{aligned} \mathcal{D}_{\mathrm{rad}} = \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) \\ + \mathcal{R}_{\mathrm{dec}} + \mathcal{R}_{\mathrm{osc}}. \end{aligned} \tag{23}$$

Therefore

$$W - P = \mathcal{B}(\rho, K) + \mathcal{R}_{\mathrm{dec}} + \mathcal{R}_{\mathrm{osc}} - \mathcal{E}_{\mathrm{ang}}, \tag{24}$$

where

$$\mathcal{B}(\rho, K) := \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{K G_1(\rho)}{4} + \frac{3}{32}\right). \tag{25}$$

Proof. Distributionally, $dw = dt - dC$. Stieltjes integration by parts, using $F(T) = 0$ and $w(0) = 0$, gives

$$\mathcal{D}_{\mathrm{rad}} = \int_0^T w(t) g(t)\, dt. \tag{26}$$

The improper boundary term at 0 vanishes because $\mathfrak{W}(t) = t^2/2$ there and $g(t) = O(t^{-1/3})$.

On $(0, \tau)$, integrate with $d\mathfrak{W} = w\, dt$:

$$\int_0^\tau wg = \mathfrak{W}(\tau) g(\tau) - \int_{(0,\tau)} \mathfrak{W}\, dg. \tag{27}$$

On (τ, T), use $w = 1/4 + \Psi'$ and $\int_\tau^T g = F(\tau)$:

$$\int_\tau^T wg = \frac{F(\tau)}{4} + \Psi(T) g(T) - \Psi(\tau) g(\tau) - \int_{(\tau,T)} \Psi\, dg. \tag{28}$$

Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, adding (27) and (28) cancels the uncontrolled interface value $\Psi(\tau)$.

Now add and subtract the extremal sawtooth values in (20). The exact identity

$$\begin{aligned} &\Psi(T) g(T) - \int_{(\tau,T)} \Psi\, dg \\ &\quad = -\frac{g(T)}{8} + \frac{3g(\tau)}{32} + \left(\Psi(T) + \frac{1}{32}\right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi\right) dg \end{aligned}$$

gives (23). By (8), $-dg$ is a positive measure on $(0,\tau)$ and dg is a positive measure on (τ, T). Equations (20), (21), and (22) therefore prove nonnegativity of both remainders. Finally substitute (9) and $\tau = KG_1(\rho)$, and then use (16). $\square$

4 A smooth lower bound for the decreasing-branch remainder

The term $\mathcal{R}_{\mathrm{dec}}$ is quantitatively important on the middle-ratio region; discarding it loses too much. It has the following elementary smooth minorant.

Lemma 3 (A global primitive minorant)**.** *For $t \geq 0$, define*

$$L(t) := \frac{t^2}{2(1+2t)}, \qquad L'(t) = \frac{t(1+t)}{(1+2t)^2}. \tag{29}$$

Then

$$0 \leq L(t) \leq \mathfrak{W}(t) \qquad (t \geq 0). \tag{30}$$

Proof. For $0 \leq t \leq 3/4$, one has $\mathfrak{W}(t) = t^2/2 \geq L(t)$. For $t \geq 3/4$, (20) gives

$$\mathfrak{W}(t) \geq \frac{t}{4} - \frac{1}{32}.$$

Moreover,

$$\frac{t}{4} - \frac{1}{32} - \frac{t^2}{2(1+2t)} = \frac{6t-1}{32(1+2t)} \geq 0.$$

This proves (30). $\square$

Define

$$\mathcal{R}_*(\rho, K) := -\int_{(0,\tau)} L(t)\, dg(t). \tag{31}$$

Because $-dg$ is positive on the decreasing branch,

$$0 \leq \mathcal{R}_* \leq \mathcal{R}_{\mathrm{dec}}. \tag{32}$$

Integration by parts is legitimate at 0 because $L(t)g(t) = O(t^{5/3})$. Hence

$$\mathcal{R}_* = -L(\tau)g(\tau) + \int_0^\tau L'(t)g(t)\, dt. \tag{33}$$

On the outer branch $t = A(z)$, one has $g(t)\, dt = -2z\, dz$. Therefore

$$\mathcal{R}_* = -\frac{\tau^2}{2(1+2\tau)} g(\tau) + \int_{\rho K}^{K} 2z\, \frac{A(z)(1+A(z))}{(1+2A(z))^2}\, dz. \tag{34}$$

This expression is smooth at every point with $0 < \rho < 1$ and $K > 0$ and has no floor or wall discontinuity.

Corollary 4 (A concrete sufficient inequality)**.** *For every $0 < \rho < 1$ and $K > 0$, define*

$$\begin{aligned}\mathcal{H}(\rho, K) := {} & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z\, \frac{A(z)(1+A(z))}{(1+2A(z))^2}\, dz, \qquad \tau = KG_1(\rho).\end{aligned} \tag{35}$$

If

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \mathcal{H}(\rho, K), \tag{36}$$

then $P(\rho, K) < W(\rho, K)$.

Proof. By (25), (34), and

$$\frac{\tau}{4} - \frac{\tau^2}{2(1+2\tau)} = \frac{\tau}{4(1+2\tau)},$$

one has $\mathcal{H} = \mathcal{B} + \mathcal{R}_*$. Now use (24), (32), and $\mathcal{R}_{\mathrm{osc}} \geq 0$. □

5 The reduced angular inequality

The preceding calculation reduces the global structural argument to the following statement on the upward-closed middle-ratio region:

$$\boxed{\sum_{\substack{n \geq 1 \\ n-1/4<T}} \big(M(R(n-1/4))^2 - R(n-1/4)^2\big) < \mathcal{H}(\rho, K)} \quad ((\rho, K) \in \mathcal{R}_{\mathrm{c}}). \tag{37}$$

This is the angular inequality passed to the companion block estimate. Everything to its right is the smooth action expression (35), and everything to its left is an explicitly centered angular rounding error. In particular, writing $r_n = m_n - 1/2 + \alpha_n$ gives its exact summands as (15); replacing them by $r_n + 1/4$ destroys the cancellation and is too coarse for the middle-ratio range.

The remaining discontinuities are transparent. For a fixed n, the summand is smooth until r_n crosses a half-integer. At $r_n = m + 1/2$, strict counting assigns $M(r_n) = m$; immediately on the side $r_n > m + 1/2$, the summand jumps upward by $2m + 1$. These are exactly the original proxy walls, now separated from the radial Stieltjes deficit.

There is a natural second summation problem behind (37). Put

$$e(r) := M(r)^2 - r^2, \qquad q(r) := -A'(r), \qquad h(t) := e(R(t)). \tag{38}$$

On every complete angular cell ($m \geq 1$),

$$\int_{m-1/2}^{m+1/2} (m^2 - r^2)\, dr = -\frac{1}{12}. \tag{39}$$

Thus the continuous comparison

$$\int_0^T h(t)\, dt = \int_0^K e(r) q(r)\, dr$$

has a favorable cell-average drift. Passing from this integral to the shifted samples $h(n - 1/4)$ produces the genuine double-sawtooth correlation. Its atoms contain the proxy-wall jumps and cannot simply be discarded by sign. In particular, if an angular wall coincides with a radial sample, Stieltjes integration by parts must retain the common-jump term dictated by the strict convention. The companion angular-block lemma pairs those jump terms with the positive quantity already retained in $\mathcal{H}$ by using the single-peaked shape of q from (6).

Remark 5 (Status). All identities and inequalities through Corollary 4 are analytic and require no finite computation. Equation (37) is the remaining angular inequality in this structural module; the companion angular-block and scalar-positivity modules prove it throughout $\mathcal{R}_{\mathrm{c}}$. Thus no upper cutoff in K is used or needed in the structural implication.